PRANAV SHINDE

Computer Vision Engineer | Inference Optimisation, Edge Deployment

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PROFESSIONAL SUMMARY

Computer Vision engineer with hands-on production experience in inference optimisation and edge deployment. Migrated YOLO segmentation models to ONNX/TensorRT INT8 (6x latency, 78% memory reduction) and profiled with Nsight Systems. Own a production CV data pipeline ingesting 500k+ images with FiftyOne deduplication and CVAT automation. Open PR (#1920) to Apple's mlx-lm. Currently at a private R&D firm (defence and government clients). AWS Certified Developer Associate.

SKILLS

Computer Vision: YOLOv8, YOLO11, YOLO26, SAM 2.1, OpenCV, PyTorch, object detection, instance segmentation, multi-object tracking (ByteTrack, BoT-SORT)

Model Optimisation: TensorRT, ONNX, INT8 quantisation, dynamic quantisation (TorchAO), Nvidia Nsight Systems, Nsight Compute, batch inference, edge deployment

Data & MLOps: FiftyOne (deduplication, embeddings, clustering), CVAT (bare-metal deployment, REST API automation), dataset curation, Docker, CI/CD

Backend & Cloud: Python, TypeScript, Node.js, REST APIs, PostgreSQL, Redis, AWS (EC2, Lambda, RDS, S3), Nginx

LLM & NLP: RAG, VLM classification (Qwen2.5-VL), text embeddings (Nomic Embed Text), prompt engineering

EXPERIENCE

Junior Engineer, Computer Vision January 2026 to Present

Badawe Engineers Pvt. Ltd – Private R&D firm, defence and government clients, Pune

- Cut inference latency about 6x (23 ms to 3.7 ms) and memory 78% (9.2 GB to 2 GB) by migrating a YOLO segmentation model from PyTorch to ONNX and TensorRT INT8; profiled the pipeline with Nvidia Nsight Systems.
- Built a SAM 2.1 boundary refiner: a YOLO model proposes bounding boxes and SAM Large 2.1 refines object edges, replacing assumed masks with edge-accurate segmentation and improving downstream model accuracy.
- Migrated CVAT from Docker to a bare-metal Linux deployment (PostgreSQL, Redis, Django, Webpack, OPA, Nginx), replaced Nuclio AI services with local Python inference bridges, and automated the full annotation workflow via REST API and CLI, saving about 3 hours of manual effort per day.
- Built a polyshape-to-polytrack converter with Watchdog-driven automation that merges frame-level shapes into persistent tracked objects, enabling video tracking without losing objects between frames.
- Trained a multi-class YOLO instance segmentation model with a custom WeightedDetectionLoss for severe class imbalance, plus P2 high-resolution detection heads to separate visually similar classes in cluttered backgrounds.
- Built a dual-GPU training and validation pipeline (RTX 5080, RTX 3060) that offloads validation to a second GPU, resolving CUDA context conflicts and cutting validation turnaround 18%.

Full Stack Intern, ML Backend January 2025 to April 2025

Vital Vistara Pvt. Ltd

- Designed AWS infrastructure for machine learning workloads (VPC, RDS, Lambda, S3, CloudWatch) and implemented CI/CD with Docker and ECS Fargate.

PROJECTS

End-to-End Computer Vision Data Pipeline

Hugging Face, Kaggle, Roboflow, FiftyOne, SAM, CVAT github.com/Py528/hf-dataset-survey

- Reduced a 500k-image dataset to 83k unique images, an 83% reduction, with FiftyOne exact and near-duplicate removal and embedding-based outlier filtering.
- Built a reproducible pipeline sourcing raw images, cleaning and filtering in FiftyOne, refining masks with SAM, and routing through CVAT to training-ready annotations; closed the loop by evaluating trained models in FiftyOne and a custom edge benchmark.

Quantised KV Cache Preallocation for Long-Context LLM Serving (mlx-lm)
github.com/ml-explore/mlx-lm/pull/1920 (PR #1920, open)

- Fixed macOS kernel panics when serving a 27B model at 64k context on 24 GB by preallocating a quantised 4-bit KV cache and enforcing single-instance cache ownership; zero crashes in a 10-turn stress test.
- Measured 2.4 to 3x faster prefill and 30 to 60% faster decode versus a comparable GGUF and llama-server setup.

Export Bench – YOLO Export-Format Benchmarking CLI
github.com/Py528/exportrace

- Built a CLI that benchmarks a YOLO model across every export format on the machine (PyTorch, ONNX, CoreML, TensorRT) and produces FPS, latency, and accuracy-delta numbers; used to verify quantisation trade-offs before edge deployment.

Bidirectional 3D Embedding Visualizer (FiftyOne Sync)
github.com/Py528/bidirectional-3d-fiftyone

- Built a self-contained 3D embedding view synced with the FiftyOne App, letting users select patch and data points in 3D space rather than a 2D embedding scatter.
- Exported 3D brain embeddings (x, y, z and patch IDs) from FiftyOne, served them via a local bridge server, and wired bidirectional lasso-selection sync: 3D to FiftyOne via Python callbacks, FiftyOne to 3D via a browser DevTools script.

EDUCATION

B.Tech in Artificial Intelligence and Data Science 2023 to 2026
Savitribai Phule Pune University (SPPU), Pune, India CGPA: 8.45

Diploma in Computer Technology 2020 to 2023
BVJNIOT, Pune, India

CERTIFICATIONS & AWARDS

AWS Certified Developer Associate, 2024

Top 10 National Finalist, Smart India Hackathon 2024 (Athlete Connect, AI sports talent assessment platform, 50,000-plus competing teams)